Real lighting and explicit chromatic controls

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Design and Implementation of a Color-Constancy-Assisted Monocular Depth Estimation System

This update responds to the supervisor's request for a clear color-constancy/depth relationship, real plus simulated evidence, expanded experiments and a method-level contribution.

Real source training	Captured real lighting	Simulation diagnostics
NYU: 368 new scenes 200 train /40 validation /128 test	ReMID: 735 real RGBs 50 captures /25 conservative labels	Vari: 264 rendered lighting RGBs Digital NYU: color/exposure controls

ReMID: captured real lighting, fixed camera and aligned reference



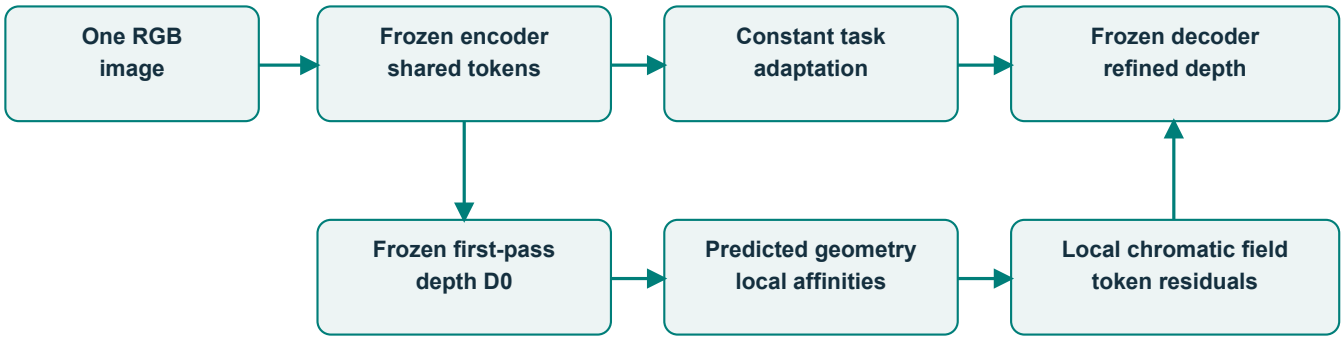
ReMID native metadata explicitly gives aligned measured depth in millimeters. We convert it to meters and keep all of ReMID external to training and checkpoint selection. Numeric scene names restart between transport shards; physical room identities are not supplied.

Completed numerical work: 15 trained variants plus the frozen model; 26,224 unique evaluated predictions across the locked test inputs, plus a 640-prediction floating-point probe. Every saved score was independently recomputed. Original experiments remain separately preserved.

The exact supervisor-mentioned work is **Point Cloud Color Constancy, CVPR 2022** (Xing, Qian et al.). It predicts illumination from measured RGB-D. The new student prototype uses its own RGB-only depth prediction as geometric context. **LumiDepth, ECCV 2026** is close prior work and supplies ReMID. These relationships constrain any novelty claim.

Main finding: the full local prototype improves digital-color AbsRel from 0.1944 to 0.1623 versus ordinary adaptation, but the local content control is slightly better at 0.1618. The validation-selected variant also transfers worse to captured ReMID than its content control. A reliable CC-specific benefit is still unproven.

Task-internal coupling and loss controls



RGB/token context also guides the chromatic field. Measured depth is supervision/evaluation only.

The local prototype predicts a relative chromatic field at encoder tokens. Normalized virtual XYZ from the frozen first-pass depth defines neighbor affinities. The field guides bounded low-rank residuals inside the depth network. The encoder and decoder remain frozen; the local module has 42,036 trainable parameters.

Depth affinity: w(i,j) = exp(-abs(log z(i) - log z(j)) /0.15). Local field smoothing and its variation penalty use the same predicted structure. These virtual coordinates are not calibrated measured world geometry.

Comparison	Question
Ordinary augmented adaptation	Does additional domain fitting explain the gain?
Brightness /generic content controls	Is the benefit explained by brightness or generic conditional capacity?
Local flat geometry	Does predicted geometry help at the same active architecture and losses?
No graph loss /no equivariance	Which objective contributes, and which hurts?

Full local objective: supervised AbsRel +0.1 paired log-depth agreement +0.2 local known-gain equivariance +0.01 geometry-weighted chromatic variation +0.001 residual regularization.

Every learned variant uses 600 paired updates, the same 200 NYU training scenes, augmentation schedule, AdamW recipe and validation rule. Checkpoints are chosen at updates 200,400,600 using only the forty NYU validation scenes. The local design was developed from global-model validation, before final test inspection.

The cue is supervised by relative digital gain changes, not calibrated absolute illuminants. FiLM, low-rank adaptation, PointNet pooling, consistency and equivariance have established prior art. The coupling is an experimentally tested research prototype; novelty is not certified by implementation alone.

All tested recipes: depth accuracy

All numbers below are mean AbsRel; lower is better. The same base checkpoint,224-pixel transform and fixed scoring masks apply. Per-image and conservative-group ReMID averages are both shown to expose weighting effects.

Method	NYU native	NYU digital color	ReMID images	ReMID groups	Vari rendered
Frozen model	0.7001	0.6968	1.7831	1.4456	1.7497
Ordinary + augmentation	0.1871	0.1944	0.6937	0.5480	0.9281
Brightness control	0.1540	0.1625	0.5427	0.4239	0.7690
Global content control	0.1539	0.1625	0.5432	0.4243	0.7698
Global chromatic, no equivariance	0.1538	0.1624	0.5440	0.4249	0.7713
Global chromatic + equivariance	0.1538	0.1623	0.5441	0.4248	0.7702
Global flat geometry	0.1539	0.1625	0.5432	0.4243	0.7700
Global predicted geometry	0.1539	0.1625	0.5433	0.4243	0.7700
Local content control	0.1531	0.1618	0.5446	0.4252	0.7732
Local flat geometry	0.1537	0.1622	0.5502	0.4271	0.7717
Local, no graph loss	0.1541	0.1626	0.5468	0.4256	0.7699
Local, no equivariance	0.1510	0.1585	0.6522	0.4848	0.8396
Local geometry + chromatic	0.1538	0.1623	0.5515	0.4276	0.7714
Local content, no equivariance	0.1540	0.1626	0.5434	0.4246	0.7703
Local flat, no equivariance	0.1518	0.1594	0.6644	0.4930	0.8485
Local, neither added loss	0.1540	0.1626	0.5478	0.4259	0.7699

Local primary versus ordinary augmented adaptation: primary 0.1623, control 0.1944; paired difference -0.0321 (interval -0.0448 to -0.0198).

Local primary versus local content control: primary 0.1623, control 0.1618; paired difference +0.0005 (interval +0.0002 to +0.0008).

Captured real data: local primary versus flat geometry: primary 0.4276, control 0.4271; paired difference +0.0004 (interval +0.0001 to +0.0009).

Validation-selected recipe versus loss-matched local content: selected recipe 0.4848, control 0.4246; paired difference +0.0602 (interval +0.0378 to +0.0848).

The validation-selected recipe is Local, no equivariance; it was chosen from NYU validation histories before final score inspection. It is distinct from the fixed full-loss hypothesis above.

The paired intervals resample whole NYU scenes or conservative ReMID labels. ReMID intervals are conditional descriptions; unknown room identities prevent a population-independence claim. This is one training seed. The prior Vari room is exploratory and does not provide independent scene generalization.

Color-specific effects and real-light stability

Controlled frozen-model probe	Changed AbsRel	Matched neutral	Paired change
Chromatic changes	0.6966	0.6935	+0.0031
Scalar exposure change	0.6999	0.6935	+0.0064

This separate frozen-model probe applies floating-point changes after fixed spatial resampling, with no subsequent eight-bit rounding. Actual encoder-input brightness drift is at most 0.000013%. The main learned-method tables preserve their original eight-bit pipeline, whose measured drift is at most 0.6284%. Native photographs, captured lighting and rendered scenes remain separate endpoints.

Paired scene-bootstrap intervals: chromatic -0.0057 to +0.0114; exposure +0.0039 to +0.0089. The chromatic interval includes zero: this probe does not establish an average color-specific error increase.

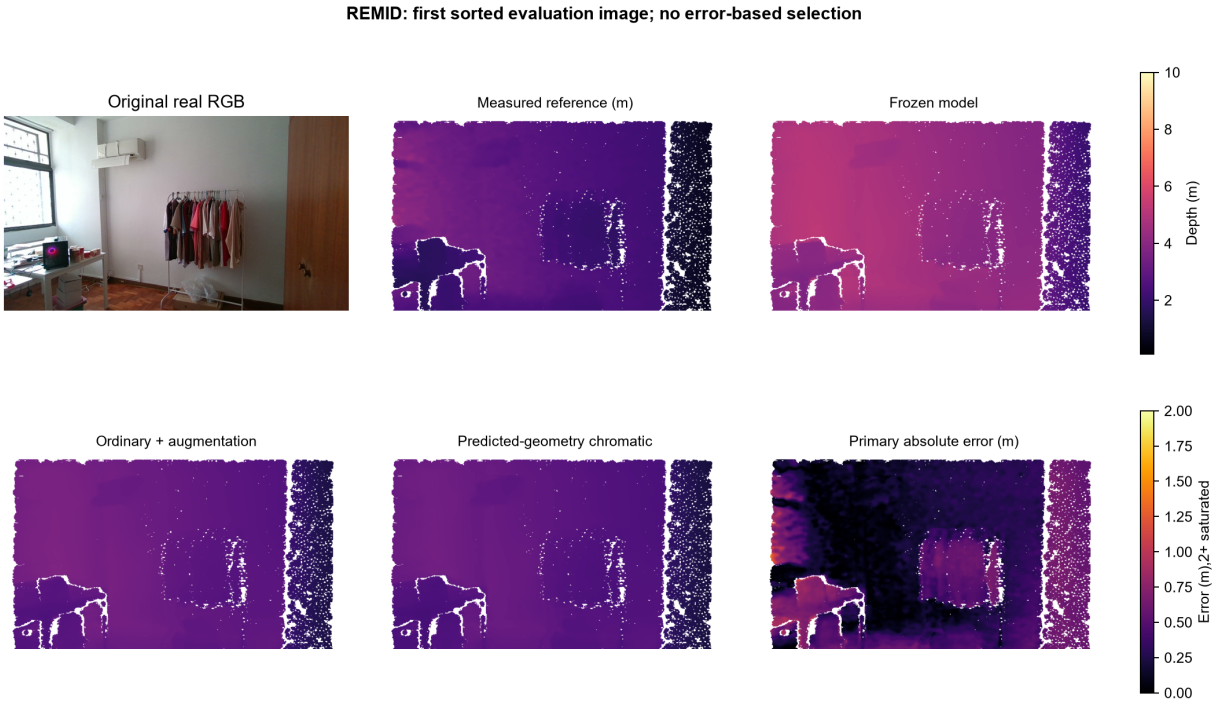
Additional chromatic sensitivity versus local content: primary 0.0076, control 0.0078; paired difference -0.000245 (interval -0.000463 to -0.000043).

The preceding difference subtracts each method's neutral-control error before comparing methods. It separates a response to the chromatic intervention from a general accuracy offset; lower differences favor the primary method.

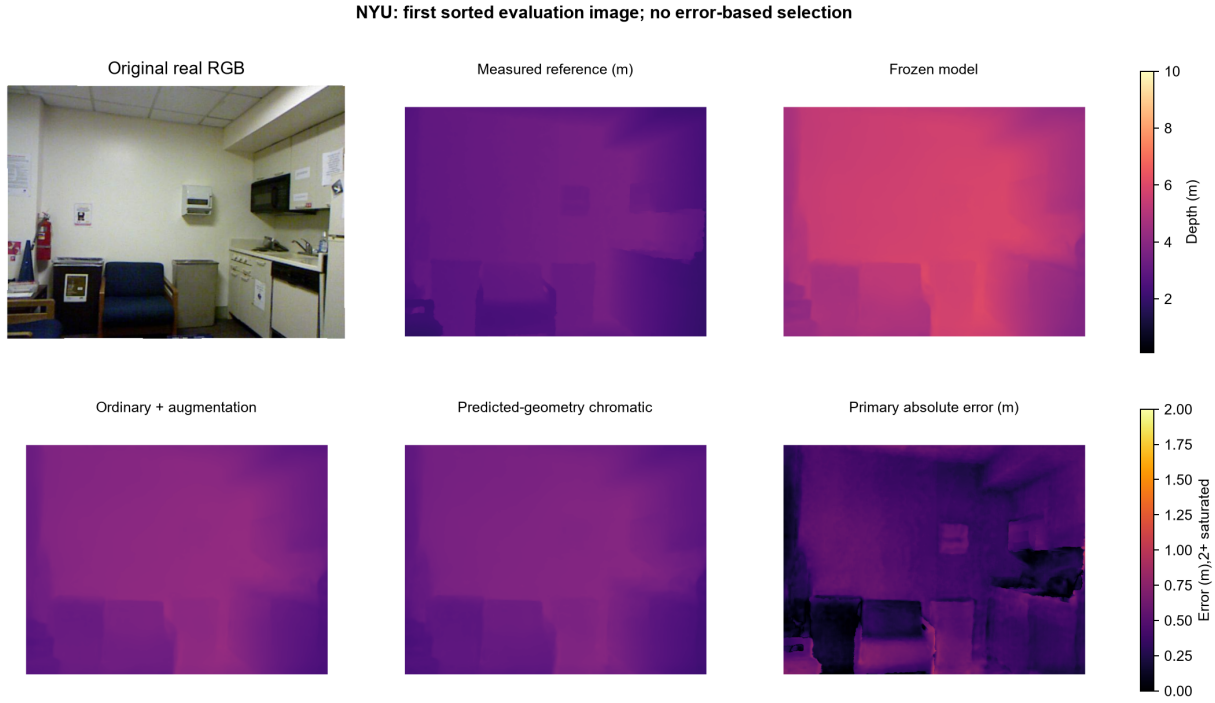
Lighting instability	Real ReMID groups	Rendered Vari views
Frozen model	0.1941	0.1025
Ordinary + augmentation	0.1316	0.0813
Global content control	0.1176	0.0742
Local content control	0.1177	0.0740
Local flat geometry	0.1196	0.0741
Local, no graph loss	0.1189	0.0742
Local, no equivariance	0.1306	0.0728
Local geometry + chromatic	0.1201	0.0742

Instability is mean absolute prediction deviation around the fixed-capture light mean, divided by shared reference depth. It is dimensionless; lower is better. Stable but inaccurate depth is not success. All 16 recipes and worst-deviation scores are available in the repository tables.

Actual saved predictions and failure evidence



Captured ReMID example: first sorted native evaluation image. The reference is aligned measured depth; zeros and out-of-range references are masked. All depth panels share 0.1-10 m; error saturates at 2 m.



New NYU example: first sorted native evaluation image. Selection uses no prediction-error ranking. Original Kinect-derived references include the authors' missing-value filling. Maps are supplied for inspection and do not replace the full numerical tables.

What remains for a paper

Evidence now available: real aligned multi-light RGB-D data, new scene-separated source data, explicit color/exposure interventions, two internal prototypes, controlled architecture/loss ablations and an independent prediction audit.

Decision from these results: do not promote the full-loss module as the winning method. The no-equivariance recipe has NYU native AbsRel 0.1510, but ReMID group AbsRel worsens to 0.4848 versus 0.4246 for loss-matched content. Small gains over flat geometry do not resolve this transfer gap. The next algorithm must address this failure rather than optimize the already-inspected test set.

Limits: absolute physical illuminant recovery has not been established. The prototype ingredients have close prior art. A gain over the frozen model alone is insufficient. Interpret the full content, brightness, geometry and loss controls together, including harms and tradeoffs.

Next research steps

1. Discuss the strongest matched-control comparisons with the supervisor and identify the remaining mechanism gap.
2. Compare competitive published methods at matched resolution and supervision, then repeat the development recipe with independent training seeds.
3. If the method changes after these results, treat this test evidence as development and reserve a new independent real dataset for confirmation.

Reproduction and code

GitHub: github.com/shirsha2002/color-constancy-depth-thesis
Read phase3/README.md, EXPERIMENT_V1.md, SPATIAL_EXPERIMENT.md and PAPER_PLAN.md. Raw data and large arrays remain local; source identities, code, trained adapters, tables and audit evidence are versioned.

Point Cloud Color Constancy (CVPR2022)
https://openaccess.thecvf.com/content/CVPR2022/html/Xing_Point_Cloud_Color_Constancy_CVPR_2022_paper.html

LumiDepth /ReMID (ECCV2026)
https://link.springer.com/chapter/10.1007/978-3-032-37314-4_9

Depth Anything V2 metric implementation
https://github.com/DepthAnything/Depth-Anything-V2/tree/main/metric_depth

NYU Depth V2 authors
https://cs.nyu.edu/~fergus/datasets/nyu_depth_v2.html

Full source and prior-art review
https://github.com/shirsha2002/color-constancy-depth-thesis/blob/main/phase3/research/LITERATURE_REVIEW.md